

(* Planet mass ratios: 1/Mass_Sun *)

massMercury = 1/6023600;

massVenus = 1/408523.5;

massEarth = 1/328900.5;

massMars = 1/3098710;

massJupiter = 1/1047.355;

massSaturn = 1/3498.5;

massUranus = 1/22869;

massNeptune = 1/19314;

General Purpose Arrays (Constants-1, Coefficients-2)

Order of the elements in the arrays is as stated below

Units for "Constant" Arrays: a [AU], e [unitless], i, Ω , ϖ , L [degrees].

Units for "Coefficient" Arrays: Coefficients for a [AU/cy], e [1/cy], i, Ω , ϖ [arcseconds/cy], L [degrees/day].

Mercury1 = [0.38709893, 0.20563069, 7.00487, 48.33167, 77.45645, 252.25084]

Mercury2 = [0.00000066, 0.00002527, -23.51, -446.30, 573.57, 4.09233880]

Venus1 = [0.72333199, 0.00677323, 3.39471, 76.68069, 131.53298, 181.97973]

Venus2 = [0.00000092, -0.00004938, -2.86, -996.89, -108.80, 1.60213047]

Earth1 = [1.00000011, 0.01671022, 0.00005, -11.26064, 102.94719, 100.46435]

Earth2 = [-0.00000005, -0.00003804, -46.94, -18228.25, 1198.28, 0.98560910]

Mars1 = [1.52366231, 0.09341233, 1.85061, 49.57854, 336.04084, 355.45332]

Mars2 = [-0.00007221, 0.00011902, -25.47, -1020.19, 1560.78, 0.52403304]

Jupiter1 = [5.20336301, 0.04839266, 1.30530, 100.55615, 14.75385, 34.40438]

Jupiter2 = [0.00060737, -0.00012880, -4.15, 1217.17, 839.93, 0.08308676]

Saturn1 = [9.53707032, 0.05415060, 2.48446, 113.71504, 92.43194, 49.94432]

Saturn2 = [-0.00301530, -0.00036762, 6.11, -1591.05, -1948.89, 0.03346063]

Uranus1 = [19.19126393, 0.04716771, 0.76986, 74.22988, 170.96424, 313.23218]

Uranus2 = [0.00152025, -0.00019150, -2.09, 1681.40, 1312.56, 0.01173129]

Neptune1 = [30.06896348, 0.00858587, 1.76917, 131.72169, 44.97135, 304.88003]

Neptune2 = [-0.00125196, 0.00002514, -3.64, -151.25, -844.43, 0.00598106]

Orbital Elements (Mathematica Style)

Order of the elements in the arrays is as stated below

Units for Arrays: **a** [AU], **e** [unitless], **i, Ω , ϖ , L** [degrees].

Mercury[T_, t_] := {0.38709893 + 0.00000066*T, 0.20563069 + 0.00002527*T, 7.00487 -
23.51/3600*T, 48.33167 - 446.30/3600*T, 77.45645 + 573.57/3600*T, 252.25084 + 4.09233880*t}

Venus[T_, t_] := {0.72333199 + 0.00000092*T, 0.00677323 - 0.00004938*T, 3.39471 - 2.86/3600*T,
76.68069 - 996.89/3600*T, 131.53298 - 108.80/3600*T, 181.97973 + 1.60213047*t}

Earth[T_, t_] := {1.00000011 - 0.00000005*T, 0.01671022 - 0.00003804*T, 0.00005 - 46.94/3600*T,
-11.26064 - 18228.25/3600*T, 102.94719 + 1198.28/3600*T, 100.46435 + 0.98560910*t}

Mars[T_, t_] := {1.52366231 - 0.00007221*T, 0.09341233 + 0.00011902*T, 1.85061 - 25.47/3600*T,
49.57854 - 1020.19/3600*T, 336.04084 + 1560.78/3600*T, 355.45332 + 0.52403304*t}

Jupiter[T_, t_] := {5.20336301 + 0.00060737*T, 0.04839266 - 0.00012880*T, 1.30530 - 4.15/3600*T,
100.55615 + 1217.17/3600*T, 14.75385 + 839.93/3600*T, 34.40438 + 0.08308676*t}

Saturn[T_, t_] := {9.53707032 - 0.00301530*T, 0.05415060 - 0.00036762*T, 2.48446 + 6.11/3600*T,
113.71504 - 1591.05/3600*T, 92.43194 - 1948.89/3600*T, 49.94432 + 0.03346063*t}

Uranus[T_, t_] := {19.19126393 + 0.00152025*T, 0.04716771 - 0.00019150*T, 0.76986 - 2.09/3600*T,
74.22988 + 1681.40/3600*T, 170.96424 + 1312.56/3600*T, 313.23218 + 0.01173129*t}

Neptune[T_, t_] := {30.06896348 - 0.00125196*T, 0.00858587 + 0.00002514*T, 1.76917 -
3.64/3600*T, 131.72169 - 151.25/3600*T, 44.97135 - 844.43/3600*T, 304.88003 + 0.00598106*t}